Digital Mohalla — Class diagram, Student 2 packages (design step A7, v1.0): Module 5 estimation, manipulation detection, decision policy and explanation

Python design classes (iteration 1), from the Algorithm doc §2–§6 and the assurance data dictionary. Each class lists the FRs it realises. Student 1 classes appear only as the interfaces Student 2 depends on (blue).

assurance (Module 5, Student 2)

boundary: Student 1 (Module 5 indirect trust)

«use»

«interface»
IAssess
assess(request: AssessmentRequest): AssessmentResult
raises EngineTimeout (2,000 ms)
FR-T-01, FR-T-07

«interface»
IOutcomeSink
record_outcome(closing_event_id: int): None {idempotent}
FR-B-15, FR-R-04

ConfigLoader
- path: str
load(): EngineConfig
NFR-D-01

«value»
EngineConfig
config_version: str
prior, half_life, interval_level
k, gamma, cap, divisors, baseline
signal weights, bands, corroboration
thresholds, value_bands, timeout_ms
NFR-D-01 (Assurance_Engine_Config_Draft.json)

«value»
AssessmentRequest
transaction_id: UUID
borrower_id, lender_id: UUID
item_id: UUID
role: Role, metric: Metric
FR-B-07, FR-T-01

«use»

«interface»
IIndirectEvidence
indirect_evidence(lender, borrower, role, metric, t_v, b): IndirectEvidence
FR-T-04 (Student 1)

«returns»

«value»
IndirectEvidence
paths: list[Path]
l, rho: float {rho ≤ 0.6}
t_indirect: float
prior_backed: bool
FR-T-04, ADR-0006 (Student 1)

AssuranceFacade
- extractor: OutcomeExtractor
- estimator: BetaPosteriorEstimator
- indirect: IIndirectEvidence
- detector: ManipulationDetector
- policy: DecisionPolicy
- explainer: ExplanationBuilder
- config: EngineConfig
assess(request): AssessmentResult
record_outcome(closing_event_id)
FR-T-01, FR-T-07, FR-B-15, NFR-D-01

OutcomeExtractor
extract(closing_event): list[OutcomeEvent]
eligible(subject, role, metric): list[OutcomeEvent]
FR-T-03 (confirmed outcomes only; disputed excluded)

BetaPosteriorEstimator
estimate(events, profile?): TrustEstimate
posterior_mean(s_w, f_w): float {shared, ADR-0006}
weight(age_days, similarity): float
FR-T-02, FR-T-03

ExplanationBuilder
build(direct, indirect, risk, decision): list[EvidenceRecord]
render(records): str
FR-T-08, NFR-U-01 (never one number)

DecisionPolicy
- thresholds: tau_mu, tau_L, n_min, rho_sup
decide(direct, indirect, risk, ctx)
: DecisionPolicyResult
rules R1–R5, R6_DISPUTE,
R6_SPARSE, R7–R9
FR-T-06, ADR-0007, ADR-0008

1 — 1 (AssuranceFacade — OutcomeExtractor)

«creates» (AssuranceFacade → AssessmentResult)

«creates» (OutcomeExtractor → OutcomeEvent)

«returns» (BetaPosteriorEstimator → TrustEstimate)

«creates» (ExplanationBuilder → EvidenceRecord)

«returns» (DecisionPolicy → DecisionPolicyResult)

«value»
AssessmentResult
state: AssuranceState
direct: TrustEstimate
indirect: IndirectEvidence
manipulation: ManipulationRiskAssessment
decision: DecisionPolicyResult
explanation: list[EvidenceRecord]
config_version: str
computed_at: datetime
FR-T-01, NFR-D-01 (stored as one assessment row)

«value»
OutcomeEvent
event_id, transaction_id
subject_id, observer_id
role: Role, metric: Metric
success: bool
status: EvidenceStatus
item_category, value_band, duration_days
occurred_at: datetime
FR-T-03 (persisted as outcome)

«value»
TrustEstimate
alpha, beta: float
mean: float
lower_90, upper_90: float
effective_sample_size: float
direct_event_count: int
FR-T-02

«value»
EvidenceRecord
factor: str
value: str
confidence: float
supporting_event_ids: list
FR-T-08

«value»
DecisionPolicyResult
decision: DecisionAction
rule_fired: str
hold_reason: HoldReason | None
escalate: bool {= hold set}
FR-T-06, ADR-0003

1 — 1..* (AssessmentResult — EvidenceRecord)

1 — 1 (AssessmentResult — DecisionPolicyResult)

1 — 0..* (AssessmentResult — SafeguardRecommendation)

«use» (DecisionPolicy → TransactionContext)

RetaliationCheck
- window_days: int = 14
is_suspected(review, records): bool
FR-T-09, ES-05

ManipulationDetector
- signals: list[SignalCheck]
- weights, bands
- corroboration: 2 signals ≥ 0.5
assess(subject, window): ManipulationRiskAssessment
FR-T-05, ADR-0007

«value»
ManipulationRiskAssessment
score: float {0..100}
level: int {1..4}
corroborated: bool
FR-T-05

«value»
TransactionContext
item_value, value_band
is_high_value, medium_value: bool
open_dispute: bool
resolved_disputes_12m: int
verification_ok: bool
request_similarity: float
FR-T-06

«value»
SafeguardRecommendation
type: SafeguardType
parameters: dict
reason: str
FR-T-01

1 — 1 (RetaliationCheck — ManipulationDetector)

1 — 1..* (ManipulationDetector — SignalCheck)

1 — 1..* (ManipulationRiskAssessment — ManipulationSignal)

«interface»
SignalCheck
name: str
weight: float
evaluate(subject, data): ManipulationSignal
FR-T-05

«value»
ManipulationSignal
name: str
available: bool
value: float {0..1}
weight: float
evidence: list[str]
FR-T-05 (evidence, never accusation)

«enumeration»
AssuranceState, DecisionAction, EvidenceStatus, SafeguardType
AssuranceState: SUFFICIENT_EVIDENCE | REVIEW_RECOMMENDED | INSUFFICIENT_EVIDENCE | ACTIVE_DISPUTE
DecisionAction: PROCEED | PROCEED_WITH_SAFEGUARDS | REQUEST_VERIFICATION | RESTRICT_HIGH_VALUE | ESCALATE
EvidenceStatus: CONFIRMED | DISPUTED | RESOLVED (Assurance DD §5)
SafeguardType: security_deposit | handover_photos | return_date_fixed | ...

ReciprocalClustering
S1, weight 0.30
evaluate(...)
FR-T-05, ES-04

ReviewSimilarity
S4, weight 0.20
evaluate(...) {char n-gram TF-IDF}
FR-T-05, ES-04, ADR-0007

Notes
- AssuranceFacade is the only entry point: Module 3 calls IAssess.assess() once per request (A8); a timeout gives an engine_failure hold, never a default state (FR-T-07).
- Trust and manipulation risk stay separate attributes of AssessmentResult and are never averaged (FR-T-06).
- Iteration 2 adds RatingVelocity (S2), NewAccountActivity (S3) and Linkage (S5) as further SignalCheck realisations.
- BetaPosteriorEstimator.posterior_mean is also the trust-edge weight (ADR-0006).
- Relationship ends: ◆ composition, ◇ aggregation, dashed open arrow = dependency, dashed hollow triangle = realisation.